The association between dietary lignans, phytoestrogen-rich foods, and fiber intake and postmenopausal breast cancer risk: a German case-control st...

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Phytoestrogens are structurally similar to estrogens and may affect breast cancer risk by mimicking estrogenic/antiestrogenic properties. In Western societies, whole grains and possibly soy foods are rich sources of phytoestrogens. A population-based case-control study in German postmenopausal women was used to evaluate the association of phytoestrogen-rich foods and dietary lignans with breast cancer risk. Dietary data were collected from 2,884 cases and 5,509 controls using a validated food-frequency questionnaire, which included additional questions phytoestrogen-rich foods. Associations were assessed using conditional logistic regression. All analyses were adjusted for relevant risk and confounding factors. Polytomous logistic regression analysis was performed to evaluate the associations by estrogen receptor (ER) status. High and low consumption of soybeans as well as of sunflower and pumpkin seeds were associated with significantly reduced breast cancer risk compared to no consumption (OR = 0.83, 95% CI = 0.70-0.97; and OR = 0.66, 95% CI = 0.77-0.97, respectively). The observed associations were not differential by ER status. No statistically significant associations were found for dietary intake of plant lignans, fiber, or the calculated enterolignans. Our results provide evidence for a reduced postmenopausal breast cancer risk associated with increased consumption of sunflower and pumpkin seeds and soybeans.